

Measuring Teacher Attrition with Labour Force Survey Data

The Case of the United States

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Why this project

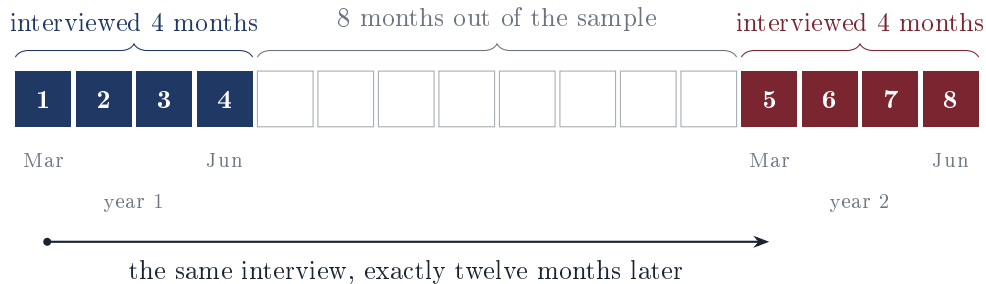
- Which of the factors we can observe correlate most strongly with the decision to leave teaching?
- Re-estimate the teacher leaving rate against the published figures of the official surveys.

The CPS replicates the official picture of the teaching workforce

	CPS	NTPS (official)
Female (%)	75.5	76.8
Master's degree or higher (%)	57.0	61.0
Average age (years)	44.0	42.9
Teachers observed	3,855	39,633

Public school teachers. CPS: supplements of 2021–2022 (school years 2020–21 and 2021–22), weighted. NTPS: National Teacher and Principal Survey 2020–21 (NCES 2022-113; Digest of Education Statistics, table 209.10; respondent count from NCES 2024-024).

Full sample: **5.0 million** person-year records, 1998–2025; **104,545** teacher observations; **48,842** teachers followed across interviews in the linked panel.



The March supplement (ASEC)

- The largest supplement of the CPS: the March sample is expanded to roughly 90,000 households.
- Every adult reports on the *previous calendar year*: longest job held, weeks worked, earnings, benefits.
- We use it to reconstruct the leaving rate.

Two ways to measure the leaving rate

Follow the person, year to year

The rotation shows the same individual twelve months apart. The CPS has no person identifier, so records are linked on fields common to both years:

household identifier

dwelling identifier

person line number within the household

validated on sex, race,

and age advancing 0–2 years

Ask about last year, each March

The supplement's retrospective question. Whoever reports teaching as last year's longest job, and no longer teaches at the March interview, left within the year.

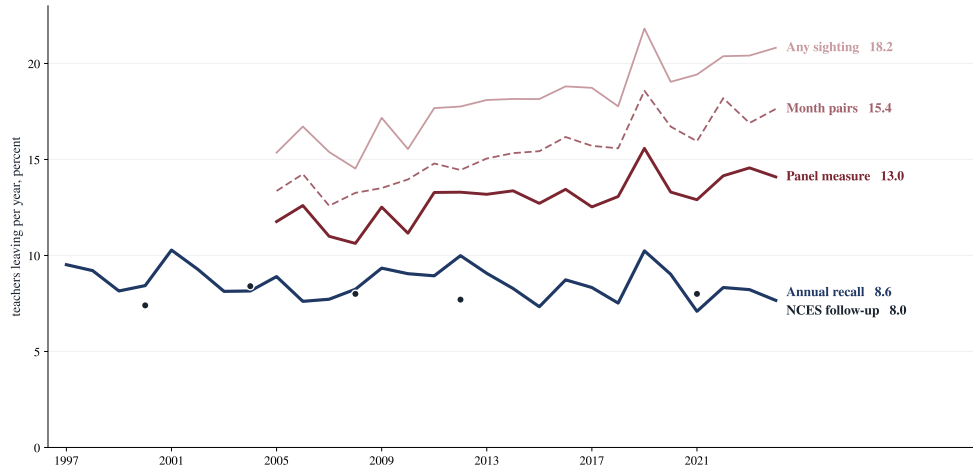
The measure used in the literature so far.

Who counts as a leaver in the linked panel?

Linking the same individuals across interview months admits many definitions. Each choice moves the rate.

- **Panel measure (benchmark).** Teaching, with a bachelor's degree, in at least two year-1 interviews; a leaver if never observed teaching in year 2, given two or more year-2 interviews, one outside the summer.
- **Month pairs.** Every teaching month is matched to its interview twelve months ahead; one person can enter several times.
- **Any sighting.** A single teaching observation in year 1 puts the person in the denominator, including one-month spells.
- **Annual recall.** The retrospective March measure, for comparison.

One dataset, one rate per definition



Dots: NCES Teacher Follow-up Survey, the official follow-up estimate.

What is the leaving rate for?

Not an end in itself: with person-level exits we can ask which observed factors correlate with the decision to leave.

Why teachers leave: what the CPS can and cannot see

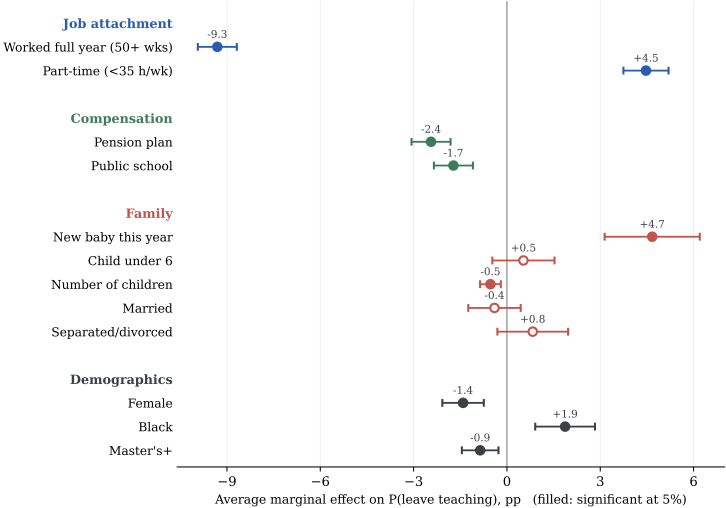
Reasons identified in the literature	Observed in the CPS?
Working conditions	no
Organizational support (mentoring, leadership)	no
Autonomy and intrinsic motives	no
Financial factors	yes
Personal life-cycle motives	yes
Sociodemographic characteristics	yes

Five broad families of reasons from our review of the literature. The CPS covers the economic and life-cycle margins.

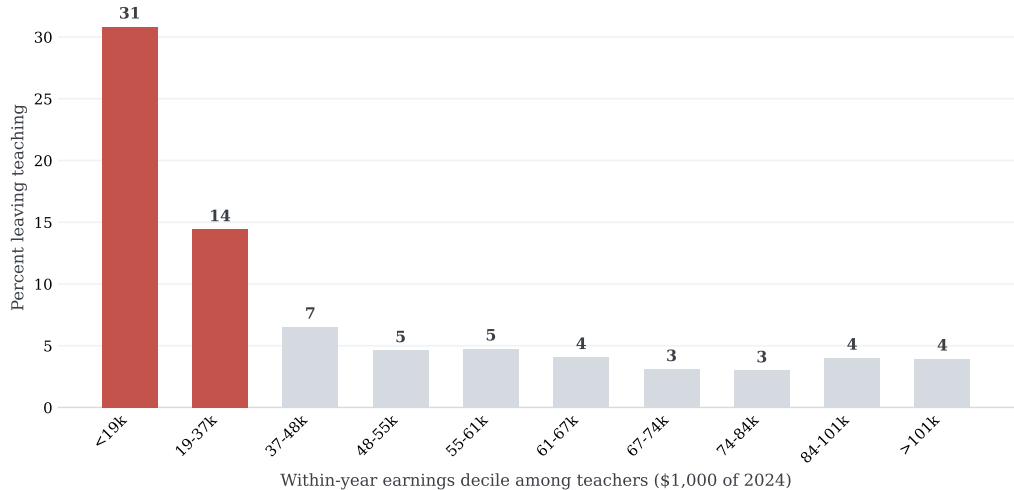
Five messages

1. **Economic and employment conditions correlate with leaving far more than demographics.**
2. Weak job attachment is the strongest single correlate: part-time and part-year teachers leave at several times the rate of full-time, full-year teachers.
3. Compensation matters: pension coverage is associated with 2.4 points less attrition, and exits concentrate in the lowest earnings deciles.
4. Life-cycle events matter: childbirth raises a woman's exit probability by about 6 points, more than in comparable professions.
5. Most leavers do not move to another job: they leave employment, and the differences across states are large.

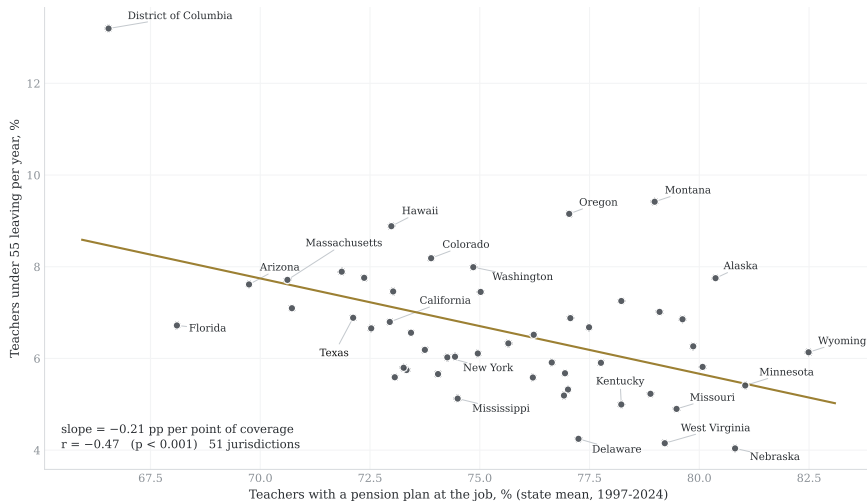
Message 1 · Economic conditions dominate demographics



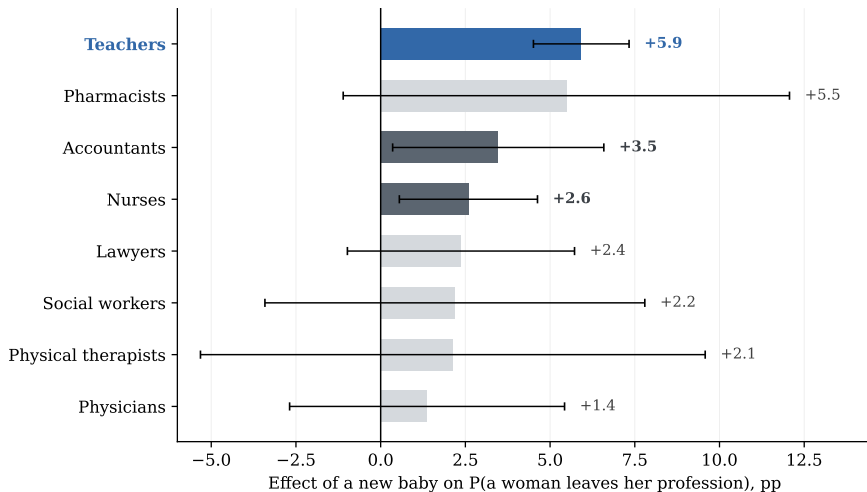
Message 2 · Exits concentrate at the bottom of the pay scale



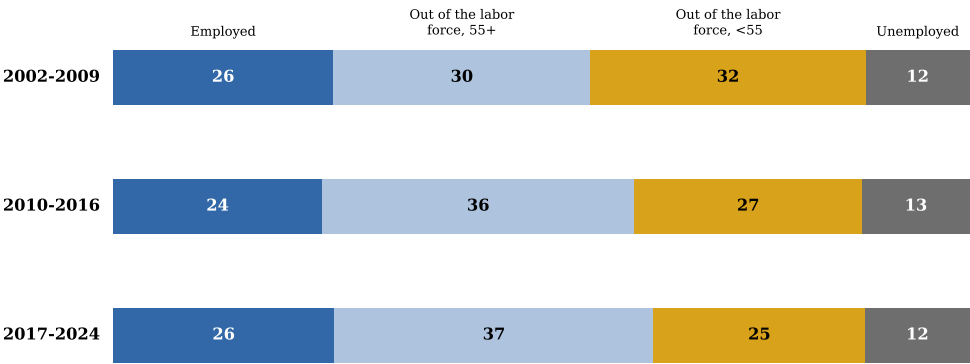
Message 2 · Where pension coverage is higher, fewer teachers leave



Message 3 · Childbirth moves teachers out more than other professionals

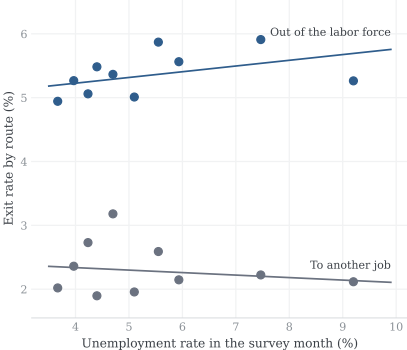
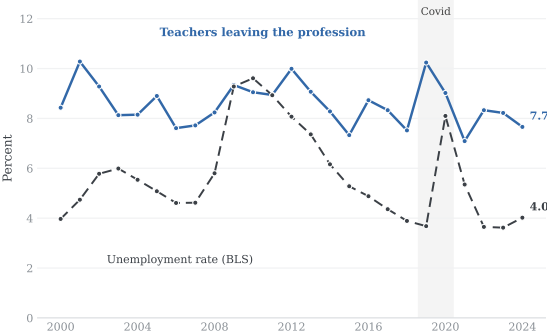


Message 4 · Most leavers exit employment, not to another job

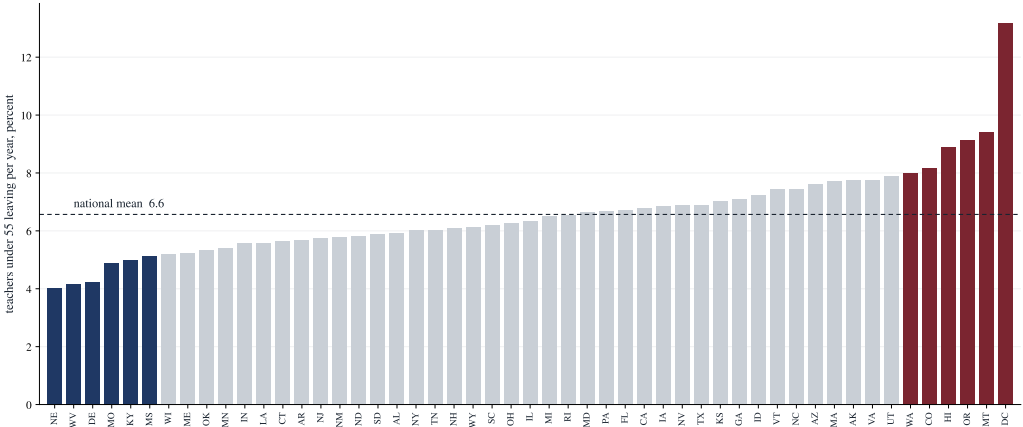


Destinations of leavers, percent of each period's exits.

Message 4 · And exits out of employment move with the cycle



Message 5 · Attrition differs widely across states



State means, 1997–2024.

Conclusion

- A labour force survey can measure teacher attrition at the person level, yearly, over three decades, and it replicates the official picture of the workforce.
- The definition of a leaver is a first-order choice: the same records yield rates from 5 to 19 percent per year.
- What the data say: attachment, pay and pensions, and life-cycle events correlate with leaving; demographics much less.
- The design travels: the same construction is feasible in the labour force surveys of Italy, Germany, the UK, France, Spain, Brazil and Mexico.